INCH-POUND

MS24182M
25 August 2021
SUPERSEDING
MS24182L
15 April 2003

DETAIL SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 100 AMPERES, 1 PST (N.O.), TYPE I, HERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

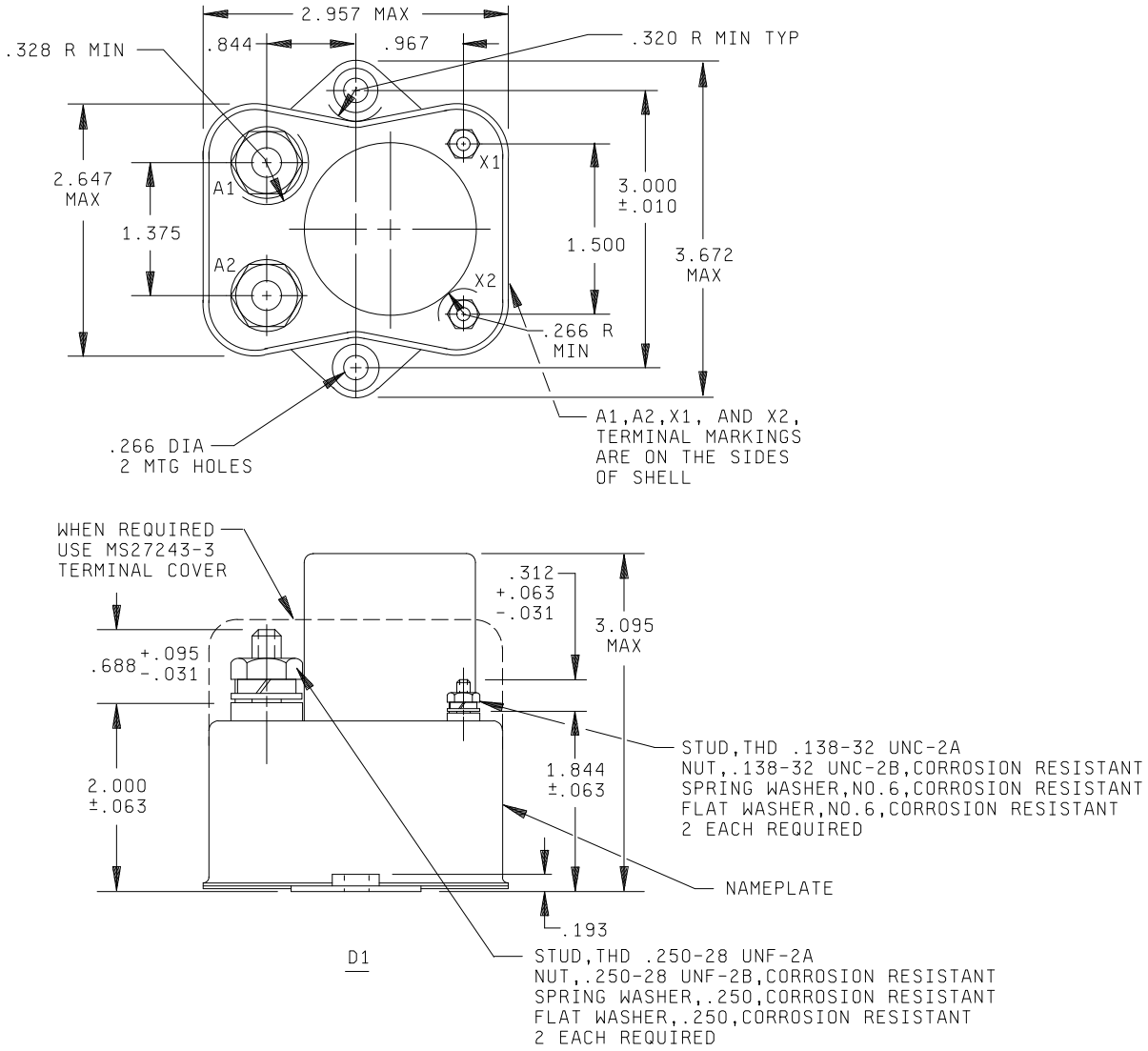
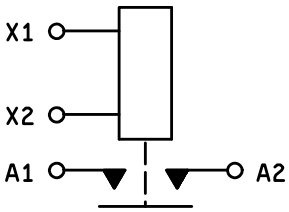


FIGURE 1. Dimensions and configurations.

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CIRCUIT DIAGRAM

Inches	mm	Inches	mm	Inches	mm
.010	0.25	.312	7.92	1.500	38.10
.031	0.79	.320	8.13	1.844	46.84
.063	1.60	.328	8.33	2.000	50.80
.094	2.39	.625	15.88	2.647	67.23
.095	2.41	.688	17.48	2.957	75.11
.138	3.51	.844	21.44	3.000	76.20
.218	5.54	.967	24.56	3.095	78.61
.240	6.10	1.375	34.93	3.672	93.27
.250	6.35	1.440	36.58	4.252	108.00
.266	6.76				

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. This specification sheet takes precedence over documents referenced herein.
5. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this specification sheet to the extent specified herein.
6. Coil and auxiliary terminals may use an additional flat washer for terminal seat.
7. Cadmium or cadmium compounds are prohibited on external hardware. A transition period to non-cadmium hardware is authorized for up to 1 year from the date of this revision.
8. Spring washer on drawing is a spring lock washer.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Dash numbers and general characteristics: See table I.

Contact data:

Load ratings: See table II.

Maximum contact drop, initial: 0.150 V.

After life test: 0.175 V.

Overload current (NO): 800 amperes.

Rupture current (NO): 1,000 amperes.

Coil data: See table III.

Duty rating: Continuous.

RFI specification: MIL-STD-461 (applicable to coil circuits of ac operated relays).

Electrical data:

Minimum insulation resistance:

Initial: 100 megohms.

After life or environmental test: 50 megohms.

Dielectric strength: Sea level, 2-5 seconds:

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1,250 V rms	1,500	1,000 V rms	1,125
Aux contacts	N/A	N/A	N/A	N/A
All other points	1,250 V rms	1,500	1,000 V rms	1,125

Dielectric strength (altitude): 1 minute.

	28 V dc	115 V ac
Coil to case	500 V rms	500
Aux contacts	N/A	N/A
All other points	500 V rms	500

Environmental characteristics:

Temperature range: -70°C to + 71°C.

Max altitude rating: 80,000 feet.

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Shock, g-level: 25 g's.

Duration: 6-9 ms.

Maximum duration contact opening: 2 ms.

Vibration, sinusoidal: See table IV.

Acceleration: 15 g's.

Part or Identifying Number (PIN): MS24182-D1.

TABLE I. Dash number and general characteristics. 1/

PIN MS24182-	Type	Coil	Terminal type	Mounting or mating socket	Auxiliary contacts	Max weight (pounds) 2/
D1	I	dc	Stud	Flange	None	1.15

TABLE II. Rated contact load (amperes per pole) case grounded. 1/

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 2/			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz
Resistive	50	100				100		5					
Inductive	10	100						5					
Motor	50	100				75							
Lamp 3/	50	50				50		.75					
Transfer load 4/													
Mechanical life reduced current	100	25						1.25					
Mixed loads	50	10				10							

1/ Absence of value indicates parameter is not applicable to this specification sheet.
2/ Absence of value indicates relay is not rated for three phase applications.
3/ The total "On" time shall be 2 seconds ±0.05 second and the "Off" time shall be 7 seconds ±2.0 seconds for a simulated lamp load.
4/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

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TABLE III. Operating characteristics.

PIN MS 24182-	Coil data											Time - milliseconds max 1/					
	Coil	Rated			Max		Max pick-up voltage			Hold voltage 1/	Drop out voltage 1/	Operate 3/	Release 4/	Contact Bounce			
														Main		Aux 5/	
		Volts 2/	Freq Hz	Ω Res +15% -10	Volts	Amp	Normal 1/	High temp test	Cont current test					NO	NC	NO	NC
D1	X1,X2	28	dc	78	29	.5	18	21	22.5	7.0	1.5	20	10	4.0	---	---	---

1/ Over the temperature range.
2/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
3/ With rated coil voltage.
4/ From rated coil voltage.
5/ Absence of value indicates parameter is not applicable to this specification.

TABLE IV. Dash numbers and general characteristics.

Part number MS24182-	5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-700 Hz	700-900 Hz	900-1,500 Hz
D1	.08 DA	.06 DA	8 g's	8 g's	5 g's	4 g's	5 g's

Qualification by similarity: See MIL-PRF-6106.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-STD-461 MS27243

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2021-024)

Review activity:
Navy - EC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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